

Research Statement: Doron Ravid

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The economics profession has long recognized that uncertainty and information have significant economic implications. Information asymmetries can lead to inefficient adverse selection (Akerlof, 1978), wasteful signaling (Spence, 1973), suboptimal contracts (Holmström, 1979), unwanted rents (Myerson, 1982), non-neutrality of money (Lucas, 1972), and more. The traditional framework for studying these distortions treats incentives and information as separate, exogenously given variables. While useful, this approach suffers from a significant drawback: information is not, in general, a fixed, given entity. Rather, information is shaped by agents' decisions as to what to learn and what to communicate. These decisions make information an endogenous variable; a function of agents' incentives. Thus, policies that are designed using the traditional toolbox of information economics may accidentally lead to changes in agents' information, which in turn could result in unintended economic consequences. My work seeks to deal with this concern by developing theoretical tools that are appropriate for studying environments with endogenous information.

Flexible Learning in Bilateral Trade

One reason information may be endogenous is that agents seek information before making their decisions. In such cases, one may want to endogenize information by modeling agents as solving an information acquisition problem. Early attempts at formulating this problem restricted the agent to choose from a fixed collection of information structures. While useful, these formulations suffer from similar issues as exogenous information models: the key force determining the shape of agents' information are the constraints the modeler imposes on the set of available signals. The rational inattention framework, pioneered by Sims (2003), addresses this concern by viewing information acquisition as a flexible decision in which agents can acquire any piece of information they want. Embracing this flexibility enables one to make

information a truly endogenous variable, determined by the economic situation at hand rather than by exogenous constraints.

A few of my papers develop the theory of flexible learning in the context of a fundamental economic settings: a seller of an indivisible good who makes a take-it-or-leave-it offer to a potential buyer. This setting is interesting both because it serves as a reasonable model of many economic situations, and because it serves as an important analytical stepping stone for studying a multitude of other economic environments. For example, by viewing a population of consumers as a single buyer with private information, one can use this setting as a model of monopolistic markets. Another example are some models of oligopolistic competition (e.g. Diamond, 1971; Varian, 1980), where each firm’s decision often takes a shape that is similar to its decision in a bi-lateral trade environment. For yet another example, one can look at the many dynamic bargaining models whose static version is equivalent to an environment in which a single seller presents a single buyer with a take-it-or-leave-it offer. In this section, I describe three of my papers that study the impact of flexible information acquisition on bi-lateral trade: ***“Ultimatum Bargaining with Rational Inattention”*** (American Economic Review 2020), ***“Learning Before Trading: On the Inefficiency of Ignoring Free Information”*** (joint with Anne-Katrin Roesler and Balazs Szentes, Journal of Political Economy 2022), and ***“Monopoly, Product Quality, and Flexible Learning”*** (joint with Jeffrey Mensch, mimeo 2022).

I begin with ***“Ultimatum Bargaining with Rational Inattention”***. In this paper, I study a mathematical model of a buyer who bargains with seller over a product of uncertain quality. The quality, $\mathbf{v}$, is determined once and for all at the beginning of the game and is observed by the seller. The seller then makes the buyer an ultimatum; that is, a take-it-or-leave-it offer. After receiving the offer, the buyer decides what information to learn about that offer and the good’s quality. The buyer’s choice is flexible: he can use any discrete signal structure whose conditional distribution over signal realizations depends on these two variables. The signal structure, the buyer’s prior, and the seller’s possibly mixed strategy determine the buyer’s attention cost. In the paper’s main model, I assume the buyer’s information costs are proportional to the expected reduction in the entropy of the buyer’s beliefs.

Once the buyer chooses her signal structure, nature draws a signal realization. Given the realization, the buyer chooses whether to accept or reject the seller’s offer.

One can interpret the paper’s setting as a model of complex bargaining situations. Indeed, complexity is a feature of many high-stakes transactions. Mergers and acquisitions involve changing the ownership of many different kinds of assets and obligations. Collective bargaining contracts set wages, benefits and work conditions for many individuals. International trade agreements determine tariffs for millions of products; and so on. For such transactions, the negotiating parties need to invest valuable resources to understand both the product and the terms of trade. They might need to carefully read long contracts, study intricate product details, or hire expensive consultants such as lawyers, bankers or engineers. In other words, they need to acquire costly information about the product’s value and the terms of trade.

To put my results in perspective, recalling a benchmark model in which the buyer perfectly observes the seller’s offer and the product’s quality is useful. In this benchmark model, there is a unique equilibrium that is characterized by two features: First, trade occurs for sure, meaning that there is no inefficiency. And, second, the seller obtains all the gains from trade.

My paper shows that adding flexible information acquisition about the seller’s product and offers reverses both features of the benchmark model. Thus, I attain two main results. First, trade must be inefficient. And second, the buyer’s surplus is strictly positive in any equilibrium where trade occurs. The paper also generates some empirical predictions. In particular, it shows that the probability of trade, the transfer given from the buyer to the seller, and the buyer’s surplus, should all increase in the quality of the seller’s product.¹

In addition to its applied results, the paper also provides a theoretical contribution to the flexible information acquisition literature. In particular, I define a new solution concept that is useful for using entropy reduction costs (and other prior-dependent cost functions) in strategic situations. My solution concept is needed because the buyer needs to pay attention to a *strategic* variable – i.e., the seller’s offers. In most of the literature, the inattentive agent pays attention to variables that are controlled

¹The paper’s ancestor, “*Bargaining with Rational Inattention*” (mimeo, 2019), shows all these features survive if the seller is allowed to make multiple offers.

by nature. These include variables determined exogenously or in aggregate, such as competitive equilibrium prices. By contrast, the buyer in my paper’s model pays attention to a strategic variable. Such variables interact in an undesirable way with the relative entropy cost function, because zero-probability events do not enter the expected entropy-reduction formula. Consequently, by threatening to receive the right signals, the buyer can deter the seller from making certain offers, which in turn results in these threats coming at zero cost. To prevent this unwanted off-path behavior, I introduce a perturbation-based refinement in the spirit of Selten’s (1975) trembling-hand equilibrium. By forcing the buyer to account for the marginal incentives of learning about zero-probability events, these perturbation prevent the buyer from making suboptimal threats off-path.

In the previously described paper, the buyer finds it costly to learn both about the seller’s product and the seller’s offers. However, there are many circumstances in which the seller’s offer takes the simple form of a posted price. In such situations, one would expect the price to be perfectly observable, whereas the product’s value should remain difficult to discern. My paper with Anne-Katrin Roesler and Balazs Szentes, ***“Learning Before Trading: On the Inefficiency of Ignoring Free Information”*** (Journal of Political Economy, 2022) studies such a model.

Specifically, we study a model where the seller has a single object for sale and has full bargaining power. Initially, the buyer does not know anything about the value of the object except its prior distribution. The buyer’s learning is modeled as flexible information acquisition, that is, she can purchase any signal about her valuation privately. Then the seller, without observing the product’s value, the buyer’s learning strategy, or the buyer’s signal realization, sets a price. Signals are costly and we assume that this cost is a smooth and strongly increasing function of the signal’s informativeness.

We interpret the above game as a stylized model of a situation where buyers have access to abundant information that allows them to learn about products prior to trading, but this learning has to occur before the term of trade are determined. For example, buyers can use online resources to learn about a mechanic’s reputation, a contractor’s ability, or an over-the-counter (OTC) asset’s value. However, to get a price quote, customers may need to bring their cars to the mechanic, have a contractor

over, or waste one's first contact with an OTC dealer. Scenarios such as these create an interaction between the seller's pricing strategy and the buyer's learning decisions. Indeed, since the buyer's willingness-to-pay depends on her information about the product, the seller's price depends on what he expects the buyer to learn. The flip side is that the seller's pricing strategy determines what information is worth learning for the buyer. For example, there may be no point of knowing more about the value of an asset if the buyer is already sure that it is below its price. So, the buyer's information acquisition depends on the seller's expected prices.

Our paper explores the above-mentioned mutual dependency between the buyer's learning strategy and the seller's pricing policy in an environment with cheap and widely accessible information. More specifically, our goal is to understand the differences between having full, free, and cheap information. As the full information case has been widely studied, we begin by analyzing the case in which learning is free. We show that, in this case, there are multiple equilibria and they can be Pareto ranked whenever the prior value-distribution satisfies the monotone hazard rate property. In the Pareto best equilibrium, which maximizes both players' payoffs across all free-learning equilibria, the buyer learns her valuation perfectly. In contrast, our main result is that, when learning is costly but the cost of information goes to zero, equilibria converge to the equilibrium that *minimizes* both players' payoffs. The difference in payoffs between the Pareto best and worst equilibria can be quite significant. For example, when the value distribution is uniform over $[0, 1]$, the seller's profit decreases from $1/4$ in the Pareto best equilibrium to approximately $1/5$ in the Pareto worst one, while the buyer's surplus drops from $1/8$ to just (slightly above) $1/25$.

Our paper's main takeaway is that possessing information might be significantly better than having cheap access to it. When information is costly, buyers must have incentives to acquire it. In equilibrium, prices fail to provide these incentives, so buyers choose to ignore large amounts of information even when costs are minuscule. In turn, this ignorance triggers prices that are too high relative to those in a full-information environment, leading to considerable welfare losses.

More broadly, our analysis warns about the danger of approximating environments with freely available data by models in which learning is costless. The reason is

that, even with easy-to-access information, learning may not be literally free, because decision makers might still need to incur attention or cognitive-processing costs when presented with new information. Hence, assuming learning has tiny but positive costs may be more accurate than assuming learning is free. Our results show that the difference between these two assumptions is not merely a manner of minutiae: equilibria of the free-learning model may be significantly better for both contracting parties than the equilibrium of the costly learning model even if learning costs are minuscule. In fact, since obtaining full information is a weakly dominant strategy, one might even assume that, under free learning, traders will coordinate on the Pareto-dominant, full-information equilibrium, whereas the Pareto-*worst* equilibrium is selected when costs are vanishingly small. As such, our work suggests that reasoning about scenarios in which information is cheap using a free-learning model may result in misleading conclusions.

In terms of applications, our model may offer a different angle for looking at markets where there is heterogeneity in traders' information. A large literature documents that the performance of traders depends on how knowledgeable they are; see, for example, Grubb (2015) and Handel and Schwartzstein (2018) for extensive surveys. Our theory might speak to those environments where such a knowledge differential is due to heterogeneity of the traders' information acquisition. To be more specific, consider an asset market that is populated by two kinds of buyers: experienced traders and novices. Experienced traders may already possess a significant amount of information related to the asset while novices are yet to learn about it. Our theory then suggests that, even if the relevant information is easy to obtain, veteran traders can get significantly better terms of trade than novices.²

In addition to its applied and conceptual takeaways, our paper also develops a new tractable class of information-cost functions. These cost functions are useful in settings such as ours where one can summarize information using distributions of posterior means (see, for example Roesler and Szentes, 2017; Dworczak and Martini, 2019). Our approach for modeling the buyer's information acquisition costs in such settings is novel in two ways. First, we define the cost of information acquisition

²Consistent with this implication, O'Hara et al. (2018) report that traders get a better price for the same bond on the same day if they traded more frequently in the past.

directly on the set of distributions of the buyer’s posterior estimate. This formulation stands in contrast to most of the literature, which models information costs as being a function of the distribution of the agent’s posterior *beliefs*, a much more complicated object than the agent’s posterior expectation. Second, most of the literature assumes costs take a “posterior-separable” form; that is, that costs are affine in the distribution of the agents’ posterior beliefs. While this class is very tractable, it is also restrictive, and stands in contrast to some available empirical evidence (Denti, 2022). We generalize this family by requiring costs to be smooth; that is, they only need to be approximable by an affine function. This assumption significantly expands the set of allowable cost functions while simultaneously preserving much of the tractability of the posterior-separable class.

The main assumption driving the results in the previously discussed paper is that the buyer’s learning must occur before she gets to see the terms of trade. However, there are many situations where buyers can consult the set of available contracts before making their learning decisions. This is the setting of my paper with Jeffrey Mensch, *“Monopoly, Product Quality, and Flexible Learning”* (mimeo, 2024). In this paper, Jeffrey and I study a model in which a seller of vertically differentiated products decides what menu to offer to a potential buyer. The buyer then sees this menu, and chooses what to learn about his type. The buyer’s information choice is flexible and costly, where costs are affine and depend only on the buyer’s distribution of posterior means. The monopolist’s menu designates a schedule of qualities and associated transfers, where the monopolist’s marginal costs are strictly increasing with quality.

When the buyer’s information is exogenous, it is well-known that the monopolist optimal allocation features “no distortion at the top” (see, e.g., Mussa and Rosen, 1978; Maskin and Riley, 1984): the type with the highest value in the distribution receives the efficient quality level. Our main result shows this feature is often reversed when the buyer gets to flexibly choose her information. Specifically, we identify conditions under which the buyer’s chosen quality *always* lies strictly below the efficient level conditional on his signal realization, including at the highest realization possible given his signal. Thus, our result suggest that accounting for the interaction between the agent’s choice of information and the principal’s menu-setting decisions

is critical for evaluating the efficiency of a given market outcome.

Credibility in Persuasion

In addition to information acquisition, incentives also influence information by changing the way agents communicate. One situation that distills this effect is one where a sender (S, she) attempts to persuade a receiver (R, he) to take a favorable action. In such a situation, S's incentives determine not only the information she wants to convey, but also the messages R is willing to believe. In other words, S's incentives impact her credibility. Credibility's importance for effective persuasion is almost self-evident and is thought to play a central role in many contexts, such as advertising, government, politics, journalism, management, and accounting. Despite its importance, credibility's exact role in optimal persuasion is not fully understood. My second major line of research aims to better understand this role.

My first paper in this line of research is ***“Cheap Talk with Transparent Motives”*** (Econometrica, 2020) with Elliot Lipnowski. In this paper, we compare the benefits S can get from communicating to R under two extreme cases: the case where S's credibility is guaranteed, captured by the famous Bayesian Persuasion model of Kamenica and Gentzkow (2011), and the case where S's messages are just cheap talk (Crawford and Sobel, 1982; Green and Stokey, 2007). In both environments, S sends a message to R, who is otherwise uninformed. After receiving S's message, R takes an action that influences both R's and S's payoffs. The difference between the two environments is the timing: whereas in cheap talk S sends the message after seeing the state, in Bayesian Persuasion S publicly commits to a messaging protocol before seeing the state.

The main difficulty in comparing these two extremes is the language of analysis. Whereas cheap talk models are usually analyzed using the tools of comparative statics, Kamenica and Gentzkow (2011) analyze their model using a “belief-based” approach, which describes every messaging protocol by its induced distribution over R's posterior belief. Using this approach, Kamenica and Gentzkow (2011) derive a geometric characterization of persuasion's limits under guaranteed credibility. More specifically, let S's *value function* be the highest value S can obtain from R responding

optimally at a given posterior belief. Then Kamenica and Gentzkow (2011) show that concavifying this function gives highest equilibrium payoff S can get in their model. By contrast, no such characterization exists for the general cheap talk model.

To overcome this issue, Elliot and I identify the “transparent motives” case—i.e., the case where S ’s preferences over R ’s actions are *state-independent*—as a case that is amenable to belief-based analysis, thereby enabling a clean comparison between cheap talk and Bayesian persuasion. In particular, we show one can use a geometric approach analogous to that of Kamenica and Gentzkow (2011) to characterize the limits of persuasion via cheap talk. In technical terms, we show that, under cheap talk, the sender’s highest equilibrium payoff is given by the *quasiconcavification* of her value function.

The geometric difference between cheap talk and commitment allows us to show that, in finite settings, almost all priors fall into one of two categories: either S can get her first-best outcome with cheap talk, or she would strictly benefit from commitment. Thus, in finite settings, the following statement holds for most prior beliefs: S benefits from commitment if and only if she can benefit from choosing R ’s beliefs in an unconstrained way.

In addition to providing a clean comparison between guaranteed credibility and cheap talk, the transparent motives specialization of the cheap talk model is interesting on its own right. Indeed, this stylized assumption fits many applications: salespeople want to sell more expensive products, politicians want to get elected, job candidates want to get hired, etc. Motivated by these applications, we develop tools for analyzing cheap talk by a sender with state-independent preferences, and demonstrate these tools by examining three examples. The first example is one in which a think tank advises a lawmaker about a reform. We identify a think tank best equilibrium which involves giving the lawmaker noisy recommendations. The lawmaker responds in kind, mixing in a way that creates a quantity-quality trade-off for the think tank: the more the think tank likes the recommended reform, the less likely is the lawmaker to implement it. We also study a broker-investor relationship, in which an investor consults his broker about an asset, and the broker earns a fee proportional to the investor’s trades. We find a Pareto-dominant equilibrium in which the broker tells the investor whether his holdings should be

above or below a fee-independent cut-off amount. Thus, the lower the broker’s fee, the better off the investor, who pays less money for the same information. Lower fees have an ambiguous effect on the broker, because they reduce her income per trade but increase equilibrium trade volume. We also conduct comparative statics in market volatility. Although higher volatility cannot hurt her, the broker strictly benefits from higher volatility only if she can effectively communicate about it to the investor. The investor’s attitude toward higher volatility is ambiguous, because it changes both the investor’s prior uncertainty and the usefulness of the broker’s information. The third example in our paper considers a multiple-goods seller who wants to maximize the probability of selling one of her many products to a buyer. In this setting, we show that the best the seller can do with cheap talk is to tell the buyer the identity of her best product, and that being able to benefit ex ante from providing the buyer with additional information about the best product is a necessary and sufficient condition for the seller to benefit from commitment.

My second paper on the topic of credibility and persuasion is *“Persuasion via Weak Institutions”* (2022, Journal of Political Economy), joint with Elliot Lipnowski, Denis Shishkin. In this paper, my coauthors and I study a parameterized middle ground between Bayesian Persuasion and Cheap Talk. To do so, we note that many senders attempt to persuade their audience via an external institution: sellers of electronic products often send their products to review websites, governments use central banks to release reports about the domestic economy, entrepreneurs rely on economic consultants to evaluate their company before an IPO, and so on. In an ideal world, such institutions would be able to credibly communicate the information chosen by the sender. In reality, such institutions often succumb to the sender’s pressures, modifying their message in the sender’s favor. Our goal in this paper is to study how an institution’s susceptibility to such pressures influences the institution’s persuasiveness and the quality of the information it provides.

Specifically, our paper studies a persuasion game between a receiver and sender (call them R and S again, respectively) where the sender’s motives are transparent as in Lipnowski and Ravid (2020). The game begins with S publicly announcing a research plan and an official reporting rule. After the announcement, S privately learns the research’s outcome and whether her reporting rule is credible. If credible, R

observes a message drawn from the announced reporting rule. Otherwise, S can freely choose the message that R sees. R then takes an action, not knowing the message's origin. We assume that reporting is credible with fixed and known probability χ , independent of any other aspect of the game, and interpret χ as the strength of S's institution. Note that when $\chi = 1$, S can never influence the report, making this setting equivalent to Bayesian Persuasion, whereas when $\chi = 0$, S can costlessly change her message ex post, and so this case reduces to a cheap talk game with transparent motives.

Our main technical result uses the aforementioned belief-based approach to characterize S's maximal equilibrium value in the intermediate credibility case, $\chi \in (0, 1)$. To do so, our theorem partitions S's equilibrium messages into two sets: messages S willingly sends when influencing the report, and messages communicated only by the official report. Given the above-mentioned results, one might guess that concavification and quasiconcavification characterize S's payoffs from official and influenced reporting, respectively. However, we show that, whereas quasiconcavification characterizes S's payoffs from influenced reporting, one cannot find S's utility from official reporting using concavification alone. The reason is that S's payoff from a message cannot surpass the utility she obtains under compromised reporting: if it did, S would have a profitable deviation. To account for this incentive constraint, one must *cap* S's value function at her utility from influenced reporting before concavifying it.

Using our geometrical characterization, we explore how the use of weaker institutions affects persuasion. One phenomenon we identify is *productive mistrust*, whereby a reduction in S's credibility results in R getting more information. This phenomenon arises when S wants to reveal intermediate information under full credibility (i.e., under $\chi = 1$). In such circumstances, a less credible S may choose to commission a report that releases more news that is bad for her, so that R believes messages that are good for S. We also show that the model can generate collapses of trust: a small decrease in S's credibility leads to large drops in S's value. More precisely, we show such a collapse occurs at some full-support prior and some credibility level if and only if S can benefit from persuasion. Finally, we examine the sensitivity of S's maximal value in the Bayesian persuasion model. We show this value is robust:

S’s value is discontinuous at $\chi = 1$ only in knife-edge specifications of the model.

The previously discussed paper follows Kamenica and Gentzkow’s (2011) lead by focusing on S’s favorite equilibrium. One way of arguing in favor of S-favorite equilibrium selection in Kamenica and Gentzkow (2011) is to assert that S’s ability to commit should give her sufficient power to steer R towards her favorite equilibrium. This argument, however, may appear less compelling in settings with a less credible sender like the one studied in “Persuasion via Weak Institutions”. In ***“Perfect Bayesian Persuasion”***, Elliot Lipnowski, Denis Shishkin, and I address this issue by asking: when is S’s equilibrium payoff unique?

To answer this question, we study the Bayesian persuasion model without imposing S-favorable equilibrium selection. In finite models, we show a sufficient condition for S’s value to be unique is that every action which is R-optimal at some belief is uniquely optimal at some other belief—a generic property. We also prove that an S-equilibrium payoff is typically unique in ordered models. Finally, we show uniqueness of S’s equilibrium payoff in the Bayesian persuasion model generates robustness to imperfect sender commitment as in the “Persuasion via Weak Institutions” paper.

New Research Agendas

The previous two sections discussed my better developed research agendas. While I intend to continue producing research on those topics, I have also begun working on two related but novel agendas aimed at developing the theory of endogenous information and uncertainty: robust predictions in flexible learning models, and models with flexible uncertainty. I currently have one paper in each of these two agendas, and intend to push them further in the near future. Below I describe these papers and relevant agendas.

Robust Predictions Under Costly Flexible Learning

As noted above, flexible information acquisition models have seen a broad array of applications (see Maćkowiak et al., 2023, for a recent review). Despite their success,

most of these models come with a caveat: they require one to make non-trivial assumptions on the information acquisition environment, and these assumptions can be challenging to verify. Consequently, one may wonder what predictions of models with flexible learning are robust to the exact specification of agents’ learning abilities.

The first new agenda I discuss in this section aims to develop a framework for making predictions that are valid across a wide range of assumptions one might make about agents’ abilities to acquire information. My current paper on the topic is *“Robust Predictions in Games with Rational Inattention”*, with Tommaso Denti. In that paper, Tommaso and I develop a framework for making such predictions in general simultaneous move games. Specifically, we consider games that begin with players covertly choosing what costly information to acquire. Subsequently, each player privately uses the acquired knowledge to take an action. Predictions are derived by solving for Nash equilibrium.

Using our framework, we identify the features that distinguish flexible information acquisition from traditional models where information is given. We demonstrate that in most settings, there is no substantive difference in terms of behavioral outcomes, yet a robust distinction emerges when considering welfare consequences. Our findings carry two key implications. Firstly, they indicate that studies that utilize flexible learning as a behavioral model need to consider and account for the subtle details of the subjects’ ability to acquire information, either implicitly or explicitly; otherwise, the analysis reverts back to standard information economics. Secondly, our paper highlights a significant array of situations where mistakenly assuming information is given, when it is actually acquired, can lead to misleading welfare conclusions. We provide tools for identifying such situations and for conducting robust welfare analysis in economies with agents who can choose what to learn in a flexible way.

The welfare difference between endogenous information and exogenous information can have concrete repercussions for policy. As a proof of concept, the paper considers a utilitarian social planner who understands the economic environment, but does not know the source of players’ information. The planner takes a worst-case approach, as is common in robust mechanism design. We characterize the set of binary-action symmetric games in which this planner’s welfare evaluations depend on whether she assumes players’ information is given or acquired.

We then apply this characterization to a regime change game, in which several investors choose whether or not to attack a distressed financial institution. We show that the planner may choose to bolster the institution’s fundamentals if she thinks information is exogenous, but not when information is acquired.

In the future, I intend to develop this agenda further, exploring the robust implications of flexible information acquisition in specific economic settings (e.g., linear-quadratic environments with Gaussian uncertainty), develop the theory to accommodate other economic settings (e.g., by accommodating dynamics), and extend our analysis to accommodate other pieces of knowledge the analyst may be aware of (such as the agents having a known level of preexisting information, and behavior in other strategic interactions).

Models with Flexible Uncertainty

Whereas learning and communication are actions that impact information, there are actions that directly change the distribution of the payoff relevant variables. There are many such examples: a high tech firm’s distribution of returns depends on its R&D decisions; the likelihood that a person gets into an accident depends on how careful she drives; the probability a patient heals depends on the doctor’s treatment decisions; and so on. Standard models studying such settings often assume a rigid relationship between an agent’s actions and the distribution of the payoff-relevant variable. The second novel agenda I intend to pursue wishes to relax this rigidity by studying models in which agents can influence the distribution of various economic variables in a flexible way.

The motivations for studying flexible uncertainty are similar to the reasons one would be interested in flexible learning. First, understanding models where agents can shape uncertainty in a flexible way may enable us to learn about scenarios where agents’ action spaces are very rich. In such scenarios, one may argue that the agent can impact the the distribution of the underlying outcomes in a flexible way, at least locally. Second, studying models with full flexibility may generate insights about the role played by the constraints imposed by classic models. And third, there are various scenarios where models with flexible production can offer significant tractability gains

compared to their rigid counterparts.

Currently, I have one completed project in this agenda, ***“Flexible Moral Hazard Paper”*** (Econometrica, 2024), joint with George Georgiadis and Balazs Szentes. In this paper, my coauthors and I study a flexible version of what is arguably one of the most basic models in information economics: the moral hazard model of Holmström (1979). Thus, we consider a principal who contracts with a single agent, with the principal’s revenue being a stochastic function of the agent’s actions. The key friction in these models is that the agent’s actions are non-contractible, and so the parties must instead contract on the output this action generates. Unlike the standard model, we assume this action space is very rich in the sense that the agent can generate *any* output distribution. Our main goal is to re-examine the validity of the main results of the standard moral hazard model.

In the standard moral hazard model, the agent’s effort is either binary or a one-dimensional set. Arguably, the most celebrated conclusion of this model is that contracts should be designed to reward the agent for outputs that are informative about the target action (see, e.g., Holmström, 1979, 2017). Because larger revenues are not necessarily more informative than smaller ones, optimal wage schemes are often non-monotone in output.³

By contrast, we show that, in flexible models, optimal wage schemes are not motivated by the informativeness of the output. Instead, they simply compensate the agent for his marginal cost of choosing the target distribution. In other words, optimal contracts are constructed so that the target distribution satisfies a generalized first-order condition: the marginal cost and marginal benefit of changing the probability of any given output must be equal. Moreover, wage schemes are always increasing in output as long as the agent’s cost of choosing a distribution is monotone in first-order stochastic dominance.

In the future, I intend to continue pushing this agenda forward by exploring the impact of flexible production and investments in other settings, such as monopolistic screening, team production, moral hazard with multi-tasking, and dynamic contracting.

³To guarantee wages are increasing, the distributions available to the agent must satisfy the monotone likelihood property.

Other Research

Each of the papers discussed above are either part of an established research agenda of mine, or an agenda I am currently in the process of developing. I also have a few papers that are not part of any specific research program. I describe these papers below.

Predictive Power of Information Costs. Compared with the traditional framework of decision-making under uncertainty, information acquisition models postulate that the agent’s behavior depends on an additional parameter: the cost of information acquisition. The appropriate values for this parameter have been the subject of intense inquiry. Some studies, such as Caplin and Dean (2015), de Oliveira et al. (2017), Dean and Neligh (2023), Dewan and Neligh (2020), and Caplin et al. (2020), seek to answer this question empirically, developing tools for identifying, measuring, and testing the agent’s cost function in experiments. Other studies take an axiomatic approach, advocating for classes of parameterized cost functions based on their characterizing features (e.g., Pomatto et al., 2023; Hébert and Woodford, 2021a; Caplin et al., 2021). An alternative string of papers explores which cost functions can be microfounded via dynamic learning (e.g., Morris and Strack, 2019; Bloedel and Zhong, 2020; Hébert and Woodford, 2021b) or as a physical cost of performing experiments (Denti et al., 2021). In *“Predicting Choices from Information Costs”* (2023, mimeo), Elliot Lipnowski and I ask a complementary question: In what ways does the agent’s information-acquisition cost restrict her behavior?

There are several reasons why one might be interested in our question. First, as explained above, multiple existing studies are dedicated to uncovering facts about information acquisition costs. By studying the restrictions these costs put on behavior, we outline the boundaries of said research program. Second, our question is relevant for applied modelers interested in understanding the connection between cost-based assumptions and behavior. For example, one of our results shows our analysis yields an immediate characterization of the set of cost functions satisfying “continuous choice,” a property which has economic implications (Morris and Yang,

2021). And third, our question can pave the path to a new way of testing whether a behavioral data set is consistent with a given cost function.

Our first result studies one’s ability to forecast the agent’s choices in a single menu using only her cost of information acquisition. We show this is a futile exercise: learning costs alone generate no testable restrictions on choice without imposing constraints on actions’ state-dependent utilities. Our second result concerns the restriction imposed by the agent’s costs of information on the way the agent behaves *across* menus. To make this point, we develop a novel class of smooth cost functions that we call *iteratively differentiable*. These are cost functions that can be locally approximated by smooth affine costs. We then show that for such costs, one can often use the agent’s actions in one menu to pin down her behavior in all submenus. Thus, once the analyst knows the agent’s cost of learning, the agent’s choice when facing the grand menu is the sole remaining degree of freedom in the agent’s behavior. Finally, we use these cross-menu restrictions for developing a test for whether or not a given agent’s decisions are consistent with a given cost function.

In addition to the above-mentioned substantive results, the paper proves some technical results on models with costly information acquisition, including: an algebraic sufficient condition for a flexible information acquisition problem to admit a unique solution; a Lagrangian first-order condition for optimality for iteratively differentiable cost functions; a reduction of the agent’s problem with differentiable costs to her problem under an affine approximation; and a generic uniqueness result for optimal behavior. Thus, our paper also provides technical tools that may be useful in future work.

Optimal Pooled Testing. A key policy tool for addressing an emerging pandemic is the creation of social distance. Although one can require the entire population to shelter in place, such a coarse policy bears enormous economic costs. An alternative solution is to use selective quarantines. Ideally, one would regularly test everyone, releasing individuals who test negative and quarantining those who test positive. Testing an entire country’s population, however, requires an unrealistically large testing capacity. One path toward relieving testing shortages is to use the same test to assess multiple individuals. Such group tests pool together samples collected from

different people, returning a positive outcome whenever one of the group’s members is infected. The idea of testing a pooled sample goes back at least as far as Dorfman (1943), who suggested it as an efficient way to screen soldiers for syphilis. Since its introduction, pooled testing has been used in various domains, including the detection of HIV (e.g., Cahoon-Young et al., 1989; Kline et al., 1989; Behets et al., 1990; Archbold et al., 1991), avian flu (e.g., Arnold et al., 2013), and rare genes (e.g., Shental et al., 2010). Some recent studies have shown pooled testing can be effective in detecting a person infected with COVID-19 in groups of 32 (Yelin et al., 2020), 48 (Shental et al., 2020), or 57 (Theagarajan, 2020) individuals. Thus, group testing holds the promise of significantly increasing a state’s testing capacity.

In **“Pooled Testing for Quarantine Decisions”** (Journal of Economic Theory, 2021), Elliot Lipnowski and I study the optimal way to use group testing to aid a country’s quarantine policy. We characterize the unique optimal allocation of tests. This allocation features *assortative batching*, whereby agents of differing infection risk are never jointly tested. Moreover, the planner tests only individuals whose prior quarantine decision is the most uncertain. Finally, individuals with higher infection risk are tested in smaller batches, because such tests minimize the informational externality of group testing.

Repeated Games. How should an informed party behave when repeatedly interacting with others who are not informed? Because information may be revealed implicitly by a player’s actions and used against the player, an informed player may sometimes wish to refrain from taking certain actions to avoid leakage of information. Alternatively, an informed player may wish to take certain actions to signal information. A rich literature, beginning with Aumann and Maschler (1966), has studied this question using the framework of repeated games with incomplete information.⁴ In Aumann and Maschler’s (1966) model, there are two patient players who play a repeated zero-sum game in which only one player is informed about the state of the world. In **“Knowing the Informed Player’s Payoffs and Simple Play in Repeated Games”** (accepted at Journal of Economic Theory), Takuma

⁴See the preface of Aumann and Maschler (1995) and the foreword in Mertens, Sorin, and Zamir (2015) for historiographical accounts.

Habu, Elliot Lipnowski, and I take the non-zero-sum version of this framework (developed by Aumann et al., 1968), and add the assumption that the informed player’s payoffs are known. Our substantive assumption is appropriate in many situations: for example, political parties want to win elections, dynasties want to remain in power, and companies want to continue to sell their goods to consumers.

Our main result is that, when the informed player’s payoffs are known, any equilibrium payoff for the informed player can be attained in an equilibrium in which the initial stages are used only for revealing information and information is revealed only in the game’s initial stages. Thus, we show that gradually leaking or signaling information has no effect on the informed player’s equilibrium payoffs whenever her motives are transparent. Our main result simplifies the computation of the informed player’s equilibrium payoffs compared to the case with general preferences, which requires the use of gradual information transmission for some payoffs.⁵

Focus Then Compare. Despite its appeal, utility maximization is at odds with a large body of experimental findings published in recent decades. To explain these findings, many researchers have suggested various forms of limited attention. One example is limited consideration, in which the agent fails to notice some of the available alternatives. Another example is limited information capacity, where the agent absorbs only some of the information available to her. A third example is salience-dependent weighing of attributes, where the agent under-weighs or over-weighs some attributes in her decision process. In the paper *“Focus, Then Compare”* (mimeo, 2019) Kai Stevenson and I suggest accommodating experimental findings via a different form of limited attention: the agent can only compare two alternatives at a time.

Specifically, we analyze a simple random choice procedure called *Focus, Then Compare* (FTC). An agent applying the procedure begins by selecting a focal option at random. The focal option is then compared sequentially to each of the other available alternatives through a sequence of pair-wise, random, and independent comparisons. The agent chooses the focal option if and only if it wins the comparison against each of the other available alternatives. Otherwise, the agent restarts the process with

⁵See, for example, Forges (1984; 1994).

replacement.

We characterize the procedure behaviorally and connect it to various experimental violations of utility maximization. We also study the extent to which one can conduct welfare comparisons using the procedure, and show that it still satisfies a version of the maxim that more options are better whenever welfare comparisons are unambiguous. As it turns out, welfare comparisons are never ambiguous when the procedure is fully deterministic, as the procedure yields a deterministic choice if and only if the choice is consistent with utility maximization. The same is not true for nearly deterministic choice: A deterministic choice rule is a limit of FTC rules if and only if it falls within the class of Componential Context Models analyzed by Tversky and Simonson (1993). As Tversky and Simonson (1993)’s model admits context effects, FTC demonstrates how even a small amounts of random mistakes can lead to significant departures from utility maximization.

Bad temptation. Temptation’s impact on behavior has been the subject of many economic studies. These studies often view temptation as pushing the agent’s decisions away from his preferences. The worst temptation, therefore, is one that steers the agent towards his least preferred alternative. In the paper “*Bad Temptation*” (Journal of Mathematical Economics, 2021), Kai Stevenson and I ask: how can one identify such bad temptation from behavior, and how does it differ from other forms of temptation?

We focus on one’s ability to differentiate bad and general temptation in a finite, static choice environment. This sets us apart from the rest of the revealed preference literature on temptation, which focuses on dynamic choices among an infinite set of possible alternatives. Thus, we avoid making any irrefutable continuity assumptions, while identifying temptation via a different channel than having a preference for commitment (i.e., a preference towards smaller future choice sets). The latter has two advantages. First, empirically, the extent to which people actually exhibit preferences for commitment is unclear. Second, unlike the existing literature, which requires the agent to be sophisticated enough to foresee his own temptation, our approach also applies when the agent is naive.

To identify temptation without sophistication, we rely on violations of the Weak

Axiom of Revealed Preference (WARP). To illustrate how temptation can lead to WARP violations, consider a dieting agent choosing an afternoon snack. The agent’s fridge contains some combination of the following three items: a healthy and non-tempting yogurt; a somewhat unhealthy and tempting sorbet; and a highly unhealthy and highly tempting ice cream. The agent always chooses yogurt when it is paired with exactly one of the other items – in these cases, the agent is willing to exert the costly self-control necessary to maintain his diet. However, resisting the very tempting ice cream is highly costly, and the agent is able to do so only by reminding himself how damaging the ice cream would be to his weight-loss regime. Therefore, when all three items are available, the agent settles on the sorbet as a form of compromise: it doesn’t require as much self-control as picking yogurt and is not as bad for his weight as choosing ice cream. This set of choices violates WARP since the agent chooses yogurt over the other two items individually but not when all three items are available.

Our main result is that, without sophistication, all temptation can be seen as bad: an agent’s choices can be rationalized by the general temptation model if and only if they can also be rationalized by a model in which the agent is tempted by her least preferred item in the menu. Moreover, we show that, for any finite number of alternatives, there is some choice behavior that can only be rationalized via bad temptation. In other words, while all temptations can be seen as bad, some temptations must be.

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